
Print your name here.

Mid-Term Examination, Business 33801:

(Monday, 5 December, 2022)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 9 questions with a total of 90 points possible. You have 90 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

I pledge my honor that I have not violated the Honor Code during this examination.

Sign your name here.

True-False-Uncertain (30 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and EXPLAIN your answer. Read each question carefully. You may take the non-italicized portion of the statement as true. **No credit will be given for answers without explanation.**

Question 1 [6 points] True, False, Uncertain? [Explain your answer.] Whisky and wine are demand substitutes and are traded in perfectly competitive markets. A barley disease has greatly reduced the yield of barley fields; barley is needed to make whisky, but not wine. *The equilibrium price of wine decreases.*

Question 2 [6 points] True, False, Uncertain? [Explain your answer.] Suppose that a competitive firm faces a price of $p = 10$ for its product. Its opportunity cost of producing output is given by $C^O(q) = 60 + \frac{1}{2}q^2$. [Note: the derivative of this function is q and 60 is an avoidable fixed cost.] *The firm should produce $q = 10$ units of output to maximize profits.*

Question 3 [12 points] True, False, Uncertain? [Explain your answer.] A firm produces wadzoozles using only zoozles and waddies as inputs, and both inputs can be adjusted in the short run. The input price of a unit of zoozle is $p_z = 18$ and the input price of a unit of waddy is $p_w = 20$.

(a). Suppose that the marginal product of a unit of zoozle is $MP_z = 20$ and the marginal product of a unit of waddy is $MP_w = 25$, at the firm's current choice of z and w . *The firm could reduce its costs by making wadzoozles using more zoozle and less waddy.*

[Note we are using w to be the number of waddies for this question; w does not represent the wage.]

(b). Suppose initially that the firm produces q_0 wadzoozles using $z = 100$ and $w = 100$ and this is the cost-minimizing way to produce this output. The price of zoozles decreases from $p_z = 18$ to $p'_z = 9$, while the price of waddies has remains unchanged. *The firm's minimum cost of producing q_0 will decrease by exactly 900.*

Question 4 [6 points] True, False, Uncertain? [Explain your answer.] Suppose that a profit-maximizing, price-taking firm is currently producing at q^* and making positive economic profits at the market price p^* . *The firm is producing at a point with economies of scale.*

Short problems (60 points total)

Question 5 [12 points] Suppose that the equilibrium price and quantity in a competitive market for instant cameras are $p^* = 200$ and $Q^* = 1000$. The slope of the demand curve (i.e., $\Delta p / \Delta Q^d$) is $-\frac{1}{2}$ and the slope of the supply curve (i.e., $\Delta p / \Delta Q^s$) is $\frac{1}{10}$.

(a). What are the elasticities of demand and supply at the equilibrium point?

(b). If the supply curve shifts up by \$1/unit at every level of Q , do market expenditures for instant cameras increase, decrease or is the outcome uncertain? Explain.

(c). If the price of smart phones (a demand substitute for instant cameras) decreases, do market expenditures on instant cameras increase, decrease or is the outcome uncertain? Explain.

Question 6 [16 points] Tyler consumes only biscotti, b , and coffee, c . Tyler's MU of b is constant at $MU_b = 12$ and his marginal utility of c is constant at $MU_c = 8$. Tyler has $I = 100$ to spend on the two goods.

(a). Suppose that $p_b = p_c = 1$. Draw Tyler's budget constraint in this setting with biscotti, b , on the horizontal axis and coffee, c , on the vertical axis. In the same figure, also draw Tyler's indifference curve map, indicating his optimal consumption of biscotti, b , and coffee, c .

[Label intercepts, axes and any points of interest clearly in your graph.]

(b). Suppose that in an effort to reduce consumption of biscotti, b , the government taxes anyone who consumes more than 50 biscotti. The effect of this policy is that for each of the first 50 biscotti, the price is $p_b = 1$ per unit, but for each one in excess of 50, the price per additional biscotti rises to $p_b = 2$. The price of coffee is always $p_c = 1$. Draw Tyler's budget constraint in this setting and his indifference curve map, indicating his optimal consumption of b and c .

[Label intercepts, axes, slopes, and any other points of interest clearly in your graph.]

Question 7 [12 points] Wednesday produces output in a perfectly competitive market. Wednesday's opportunity cost of production is given by

$$C^0(q) = 20 + 2q + \frac{1}{2}q^2.$$

[Note: the derivative of this function is $2 + q$ and 20 is an avoidable fixed cost.]

(a). If Wednesday can sell as much as she likes at the market price of $p = 10$, how much should she produce?

(b). Suppose that Wednesday has written some forward contracts with a subset of consumers that allow her to sell up to $q = 4$ units at a price of $p = 20$. In addition, suppose that spot market price for all other consumers is $p = 12$. Assume that Wednesday can sell as many units as she wants on the spot market in addition to producing up to 4 units for the forward contracts. How much should Wednesday supply to the spot market and how much should she supply on the forward contracts? Explain.

Question 8 [10 points] Xavier currently chooses to work overtime each weekend for an additional \$800 per weekend. Xavier doesn't mind doing the overtime because he has no social life and his free time has zero value to him. Next weekend, however, he has been offered an opportunity to consult on a simple project that will require 20 hours of work to research and write the report. He can do the entire consulting project over one weekend or he can work overtime at his original job that weekend; he cannot do both.

Xavier also needs to purchase some data to do this consulting project, and the data will cost \$1000. That said, his friend Bianca would pay \$400 to have access to the data once he is finished using it. (The seller of the data allows such transfers of the data.) Lastly, his overtime

salary is taxed by the government at 40 percent, while the consulting project revenue is tax free because of a special small-business tax incentives program.

- (a). What is Xavier's opportunity cost of doing the consulting project?
- (b). If Xavier is offered \$1500 to complete the project, should he accept it?

Question 9 [10 points] Enid (who has no wealth in the bank) is offered a choice between (1) a payment of \$900 and (2) a 80 percent chance of \$2500 and a 20 percent chance of nothing.

- (a). If Enid's utility function from money is $u(x) = \sqrt{x}$, which should she choose – the riskless payment or the gamble? Explain.
- (b). Suppose that Enid has the risky portfolio – an 80 percent chance of \$2500 and a 20 percent chance of nothing. What is the risk premium she associates with this portfolio?

Answers to mid-term examination

1 False/Uncertain. The reduced yields for barley implies that the supply of whisky will shift inwards. This has the effect of increasing the equilibrium price of whisky. Because wine and whisky are demand substitutes, the increase in the price of whisky has the effect of shifting the demand curve for wine outwards. With an outward shift in the demand for wine, the equilibrium price (and quantity) of wine will increase, not decrease.

2 False. The marginal output rule tells us that the optimal production (if it is positive) should satisfy $MC(q) = q$ equals $MR = p$, so $q = 10$. The shutdown rule, however, says to set $q^* = 0$ if economic profit is negative at 10. Economic profit at $q = 10$ is $\pi = p \cdot q - C^O(q) = 10(10) - 60 - \frac{1}{2}(10^2) = 100 - 60 - 50 = -10 < 0$. Hence, the firm should shutdown and produce nothing.

An alternative, but much harder way to arrive at the same conclusion is to solve for the minimum AC^O for the firm. This minimum occurs where $MC(\hat{q}) = AC^O(\hat{q})$, and thus

$$\hat{q} = \frac{60}{\hat{q}} + \frac{1}{2}\hat{q}.$$

Simplifying, we find that $\hat{q} = \sqrt{120} \approx 10.95$. At that value, we have minimum $AC^O = \sqrt{120} = 10.95$, which exceeds $p = 10$, so the firm should shut down and choose $q^* = 0$.

3 (a). False/Uncertain. The output produced by spending \$1 more on zoozle is $\frac{MP_z}{p_z} = \frac{20}{18} = \frac{10}{9} = 1.11$. The output produced by spending \$1 more on waddy is $\frac{MP_w}{p_w} = \frac{25}{20} = \frac{5}{4} = 1.25 > \frac{10}{9}$. Assuming that isoquants are smooth (the regular case) the firm could reduce its costs by shifting its inputs from zoozles to waddies. (Alternatively, you could have noted $MRTS_{z,w} = \frac{20}{25} = 0.80 < \frac{p_z}{p_w} = \frac{18}{20} = 0.90$

(b). False/Uncertain. If the firm continues to produce q_0 using the same mixture of 100 units of zoozle and 100 units of waddy, then the firm's cost will decrease by $\Delta p_z \cdot 100 = (18 - 9)100 = 900$. In general, however, when the input price ratio changes, the firm will find it optimal to choose a different mixture of inputs. If that is the case (the regular case), then the firm will lower costs by more than 900 at it chooses a better input mix.

4 False. The marginal output rule requires $p^* = MC(q^*)$. The fact that the firm is making economic profits implies that $p^*q^* > C^O(q^*)$, which is equivalent to the statement that $p^* > C^O(q^*)/q^* = AC^O(q^*)$. Putting these two relationships together, we have

$$p^* = MC(q^*) > AC^O(q^*).$$

Thus, a price taking firm making positive economic profits will generally have $MC(q^*) > AC(q^*)$ at the optimal point of production, and hence there are diseconomies of scale (increasing AC).

5 (a). It is critical that you consider the slopes properly when apply the formulae:

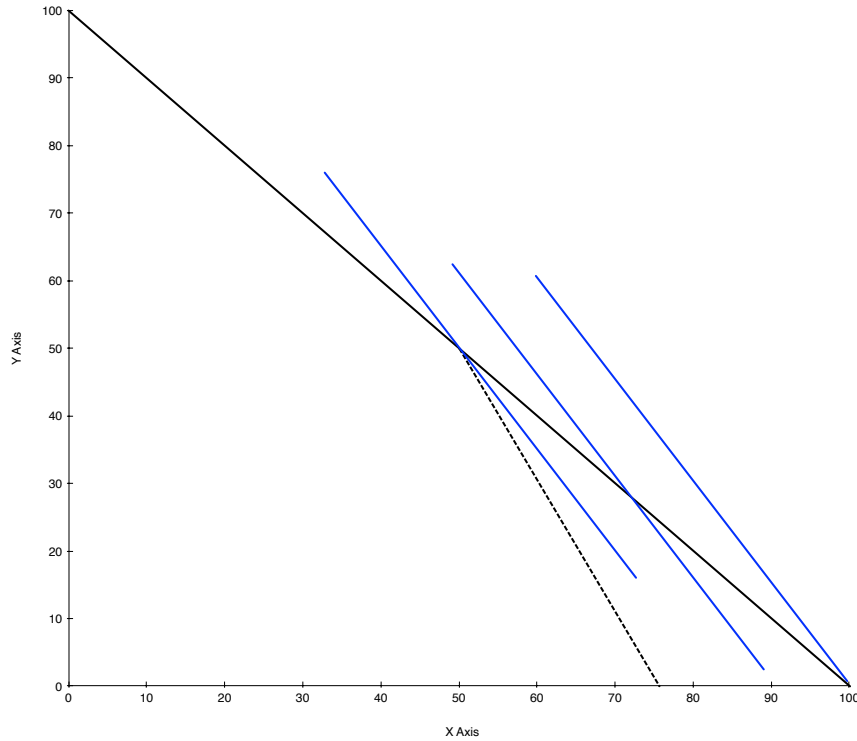
$$\varepsilon^d = \frac{\Delta Q^d}{\Delta p} \frac{p^*}{Q^*} = -2 \frac{200}{1000} = -400/1000 = -0.4$$

$$\varepsilon^s = \frac{\Delta Q^s}{\Delta p} \frac{p^*}{Q^*} = 10 \frac{200}{1000} = 2000/1000 = 2.$$

(b). Given the supply curve shifts, the equilibrium points (before and after) are both on the demand curve and the change is represented by *an increase in price along the demand curve*. In general, the effect on market expenditures is ambiguous because p increases and Q decreases. Given that demand is inelastic from part (a), however, we know that an increase in price along the demand curve in the inelastic region will increase market expenditures.

(c). Because the price of a demand substitute decreases, the demand curve shifts inward. It follows that the equilibrium market price decreases and the equilibrium market quantity decreases. Hence, market expenditures must decrease because both the width and height of the revenue rectangle decrease.

6 Figure for (a)-(b):



(a).

$$MRS_{b,k} = \frac{MU_b}{MU_k} = \frac{12}{8} = 1.5.$$

Because this is constant, Tyler's indifference curve map is a set of parallel indifference curves, each with the slope of -1.5. The blue lines represent Tyler's indifference curve map. The solid black line represents his budget constraint. Given that Tyler's $MRS_{b,c}$ exceeds 1 (the price ratio of the budget line), the optimal point is at $b^* = \frac{I}{p_b} = 100$ and $c^* = 0$.

(b). With the imposition of the tax, the price of b increases from 1 to 2 in the region $50 \leq x \leq 100$. The dotted line represents this new segment on the budget constraint. Tyler can now consume anything below the minimum of the original budget line and the dotted line. Importantly, the left segment of the budget line has price ratio 1 and the right segment has a price ratio of 2. Because $MRS_{b,c} = 1.5$ and falls between those two numbers the optimal point will be at the intersection, $b^* = 50$, $c^* = 50$.

7 (a). The marginal output rule requires that q^* satisfies $p = MC(q^*)$ if the firm produces any output. Thus, $10 = 2 + q$. The result is that Wednesday should produce at $q = 8$ if the firm decides to produce. To determine if operation is optimal, note that at $q^* = 8$, revenues are $p \cdot q = 10 \cdot 8 = 80$, while costs are $C^O(8) = 20 + 2(8) + \frac{1}{2}(8)^2 = 68$. Thus, the firm should operate.

(b). With the forward market offering 20 per unit, Wednesday should supply her first 4 units to that segment of the market. Hence, $MR = 20$ for the first four units. For units beyond the four allocated to the forward contracts, she can allocate them to the spot market at $p = 12$. Her marginal cost of producing x for the spot market is $C'(4 + x) = 2 + 4 + x = 6 + x$. This is set equal to 12 to obtain $x^* = 6$. Hence, if Wednesday operates, she should supply a total of 10 units: 4 units on the forward contract and 6 units to the spot market. This would generate revenues of $R = 20(4) + 12(6) = 152$. Wednesday's costs of producing 10 units is $C(10) = 20 + 2(10) + \frac{1}{2}(10)^2 = 130$. Because this is less than $R = 152$, Wednesday should operate as indicated above and produce $q^* = 10$. Alternatively, you could also have noted it was optimal to operate in (a) with lower prices, therefore Wednesday would want to operate under the facts of (b) where the prices are higher.

8 (a). Xavier uses up two resources in doing the project: (1) the value of lost overtime and (2) net cost of data. The opportunity cost of the lost weekend is \$800 times $(1-.40) = \$480$, which gives the lost after-tax consumption from giving up overtime for the weekend. The net opportunity cost of the data is the purchase price of \$1000 less the sales price of \$400 = \$600. Thus, the opportunity cost of doing the consulting project is \$480+\$600=\$1080.

(b). Given the opportunity cost of the consulting project is \$1080 and the benefit is \$1500, Xavier should take on the consulting project.

9 (a). Enid's EU from the 900 payment is $EU = \sqrt{900} = 30$ utils. Enid's EU from the gamble is

$$EU = 0.8\sqrt{2500} + 0.2\sqrt{0} = 0.8(50) = 40.$$

Hence, Enid should take the risky gamble.

(b). From (a), Enid's certainty equivalent of the risky payment satisfies

$$\sqrt{CE} = 40,$$

so $CE = (40)^2 = 1600$. The expected payoff of the risky portfolio is $E[\tilde{x}] = 0.8(2500) + 0.2(0) = 2000$. The difference is

$$\rho = 2000 - 1600 = 400,$$

which is Enid's risk premium.